Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery – AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

R26-DS-010

Project Proposal Report

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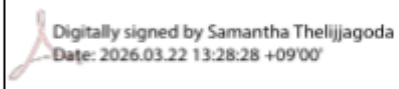
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

In today's healthcare landscape, caregivers are facing an invisible yet critical crisis which is burnout. Despite rapid technological advancements that have improved diagnostics, treatment, and administrative efficiency, the well-being of care givers is often overlooked. Long working hours, emotional strain, staff shortages, and high-pressure environments continue to push caregivers beyond their limits, affecting both their health and the quality of patient care.

This research introduces a smart, AI-driven system specifically designed to support caregivers by addressing the root causes of burnout. Unlike traditional management tools that primarily focus on task tracking and scheduling, this solution leverages advanced data science techniques to understand the human dynamics of the workplace. It predicts when a caregiver is approaching burnout and identifies contributing factors such as workload imbalance, shift patterns, and team interaction inefficiencies.

By analyzing complex patterns in task distribution and collaboration networks through machine learning and graph-based modeling, the system provides real-time, data-driven recommendations. These include optimized workload distribution, better shift planning, and improved team coordination. Interactive dashboards translate this complex data into clear, actionable insights for supervisors and administrators, enabling timely interventions.

Ultimately, the goal of this system is to foster a healthier, more supportive work environment where caregivers can maintain their well-being while delivering high-quality care. By prioritizing the mental and physical health of healthcare workers, this approach not only improves job satisfaction and retention but also enhances overall patient outcomes. In doing so, it contributes to building a more resilient and sustainable healthcare system for the future.

Keywords: Caregiver Burnout, AI-Assisted Workload, Workforce Deterioration, Predictive Analytics, Healthcare Optimization.

ACKNOWLEDGEMENT

The development of this research proposal has been both a challenging and rewarding endeavor. I would like to express my sincere appreciation to all individuals who have contributed their expertise and support during this initial stage of my study.

First and foremost, I extend my deepest gratitude to my main supervisor, Prof. Samantha Thelijjagoda, for their invaluable guidance, insightful feedback, and continuous support in shaping the core objectives of this research. Their expertise has been instrumental in refining the technical direction and overall quality of this proposal.

I would also like to convey my sincere thanks to my co-supervisor, Mr. Samadhi Rathnayake for their time and constructive feedback, which have significantly enhanced the feasibility and clarity of the proposed system.

Furthermore, I am grateful to the Lecture-in-Charge for this module, Prof. Koliya Pulasinghe for providing a well-structured academic framework and clear guidelines that directed the preparation of this document. Their guidance has been essential in ensuring that this proposal meets the required academic standards.

Finally, I wish to acknowledge my peers and family for their unwavering encouragement and support. Although this research is still in its preliminary stages, their motivation has given me the confidence to progress toward the implementation phase.

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LIST OF ABBREVIATIONS

AI - Artificial Intelligence
ML - Machine Learning
DSR - Design Science Research
HER - Electronic Health Record
MBI - Maslach Burnout Inventory
ZBI - Zarit Burden Interview
API - Application Programming Interface
UI - User Interface
UX - User Experience
SaaS - Software as a Service
AWS - Amazon Web Services
GCP - Google Cloud Platform
WHO - World Health Organization
CDC - Centers for Disease Control and Prevention
SDG - Sustainable Development Goal
CEAI - Centre of Excellence for Artificial Intelligence
SMART - Specific, Measurable, Achievable, Relevant, Time-bound

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1. INTRODUCTION

1.1. Background and Context

Caregiver workforce deterioration has emerged as a critical challenge in modern healthcare systems, particularly due to increasing workload demands, staffing shortages, and complex care environments. Caregivers are responsible not only for direct patient care but also for administrative documentation, coordination, and communication tasks, all of which contribute to cumulative workload pressure. These pressures do not manifest immediate burnout but rather develop gradually through repeated exposure to high-demand conditions such as long shifts, emergency situations, and inadequate recovery time. This gradual deterioration often remains undetected until it leads to serious outcomes such as reduced care quality, increased errors, or workforce attrition [1].

Factor Category	Description	Real-World Example
Workload Intensity	High number of patients/tasks per shift	One caregiver handling 10+ patients
Schedule Instability	Irregular or rotating shifts	Day-night shift switching
Emergency Exposure	Frequent exposure to critical situations	ICU or emergency ward work
Recovery Deficit	Lack of rest between shifts	< 8 hours rest between shifts
Administrative Load	Documentation and EHR-related tasks	Long hours spent on records
Team Imbalance	Unequal workload distribution among staff	Few staff handling most workload

Table 1: Factors Contributing to Caregiver Deterioration

The operational needs of caregiving environments create challenges which require organizations to allocate work tasks and create staff schedules because these decisions affect the health of their caregivers. Although recent advances in AI have enabled burnout prediction and workforce analytics using operational data, these systems are still limited in their ability to model caregiver deterioration as a dynamic, multi-dimensional process. Most approaches rely on risk scoring or retrospective dashboards, and do not fully capture temporal workload patterns, relational team structures, or support actionable intervention such as workload redistribution. As a result, healthcare organizations lack proactive mechanisms to identify early warning signals and intervene before workforce instability escalates into burnout or turnover [2].

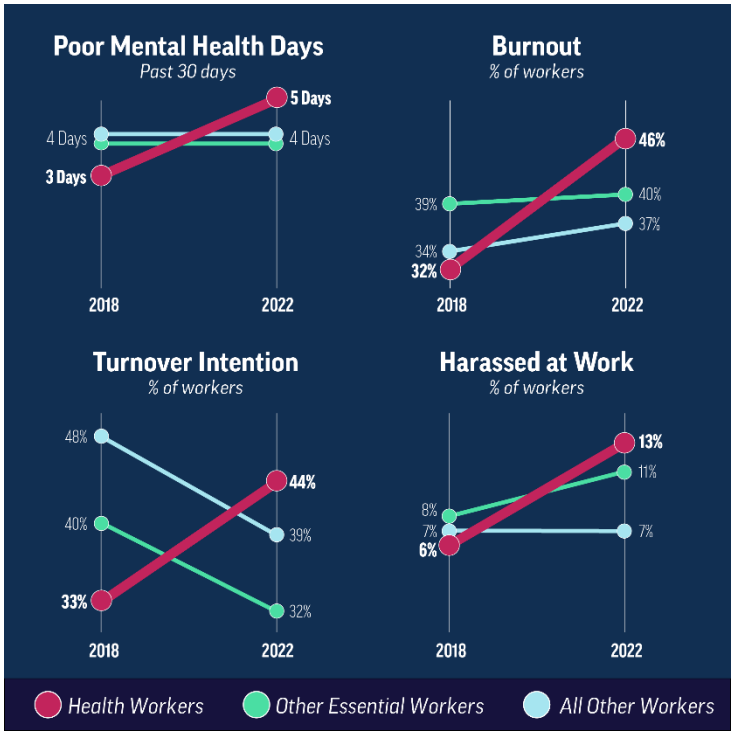


Figure 1: Nurse Burnout And Workload Stress Visualization From U.S. Centers For Disease Control And Prevention from https://www.cdc.gov/vitalsigns/health-worker-mental-health/index.html

The primary individuals affected by caregiver workforce deterioration are caregivers and nurses, who experience increasing physical and psychological strain. However, this effect does not stop with the employee, as it has been shown to be significantly associated with decreased patient safety, quality of care, and patient satisfaction, thus becoming a societal issue as opposed to an occupational one [3].

Healthcare institutions and administrators are also significantly affected, as workforce deterioration leads to increased absenteeism, reduced productivity, and higher turnover rates. This creates a cycle of staffing shortages, further increasing workload pressure on remaining staff. At a broader level, families and communities are impacted through reduced care reliability and accessibility. Therefore, addressing caregiver workforce deterioration is not only important for individual wellbeing but also essential for maintaining healthcare system sustainability and performance [4].

Recent global statistics highlight the severity of caregiver workforce challenges. According to the CDC Vital Signs report, approximately 46% of healthcare workers reported feeling burned out often or very often in 2022, reflecting a significant increase compared to previous years [5]. Additionally, the WHO estimates a global shortage of 11 million health workers by 2030, indicating a growing gap between healthcare demand and workforce capacity [2]. These statistics demonstrate that caregiver workload and burnout are not localized issues but global concerns affecting healthcare systems worldwide.

In addition to burnout statistics, there is also research suggesting that operational factors, including staffing and workload distribution, have a direct impact on patient outcomes. For example, there is evidence to suggest that staffing levels contribute to patient safety issues, including falls and delayed care interventions [6]. These findings emphasize the need for

intelligent systems capable of analyzing workload patterns and supporting proactive workforce management strategies.

Existing solutions for addressing caregiver workload and burnout can be broadly categorized into standardized assessment tools, healthcare workforce management systems, and AI-based predictive models. Standardized instruments such as the Maslach Burnout Inventory (MBI) and the Zarit Burden Interview (ZBI) are widely used in both clinical and research settings to quantify burnout and caregiver burden. For example, MBI has been extensively applied in healthcare workforce studies to measure emotional exhaustion, depersonalization, and reduced personal accomplishment [7], while ZBI is commonly used to assess caregiver burden through self-reported questionnaires [8]. However, these tools rely on periodic self-reported data, which introduces subjectivity and recall bias, and they fail to capture continuous, evolving workload dynamics in real-world caregiving environments.

In operational settings, hospitals employ Electronic Health Record (EHR)-integrated dashboards and workforce analytics systems, such as those implemented in Epic Systems and Cerner Millennium platforms, which provide real-time information on staffing levels, patient assignments, and workload indicators. These systems enable monitoring of operational metrics such as hours worked, overtime, and patient-to-staff ratios [9]. However, they primarily offer descriptive and retrospective insights, lacking predictive capabilities and failing to model how workload conditions evolve over time.

Recent research has explored the use of machine learning models for burnout prediction, leveraging data such as survey responses, EHR usage logs, and demographic information. For instance, Linzer et al. developed predictive models using healthcare workforce survey data to identify burnout risk among physicians [11]. Similarly, the HiPAL framework uses deep learning on large-scale EHR activity logs to model clinician behavior and predict burnout-related outcomes [12]. While these approaches represent significant advancements, they exhibit key limitations. Most models treat caregivers as independent units, ignoring relational dependencies such as team structure and shared workload environments. Additionally, many models rely on snapshot or cross-sectional data, limiting their ability to capture temporal progression of deterioration.

Furthermore, existing AI-based solutions typically focus on prediction rather than intervention. Although some workforce optimization systems address scheduling, such as nurse rostering models using constraint optimization techniques [13], they operate independently from burnout detection systems and do not incorporate behavioral or temporal risk indicators. This fragmentation results in a lack of integrated decision-support systems that can connect detection, explanation, and actionable intervention.

To address the identified limitations, this research proposes a working web-based application component within a broader multimodal computational intelligence system, focusing specifically on caregiver workforce deterioration detection and adaptive workload redistribution. The system is designed to collect and process operational, behavioral, and environmental data related to caregiver activities through a centralized platform. By integrating multiple data sources, the application provides a comprehensive and continuous view of

caregiver workload conditions. Unlike existing systems that rely on static measurements, this component models caregiver deterioration as a dynamic and evolving process, capturing how workload pressure accumulates and changes over time.

The interactive web application which serves as the proposed solution consists of multiple essential components. The system comprises three main components which include a personalized behavioral profiling module that creates a dynamic baseline to assess each caregiver and a temporal analysis module that identifies patterns and trends which emerge during shifts and a risk accumulation model which monitors the progression of workload stress throughout various intervals. The system includes a graph-based relational analysis module which enables users to study team relationships and their shift patterns and workload distribution among caregivers. The system provides an explainable decision-support interface which allows supervisors to comprehend detected risks through straightforward and understandable insights. The application enables adaptive workload redistribution because it detects which caregivers are working beyond their capacity and proposes equitable methods to reassign their tasks. The proposed solution develops a usable and expandable tool for enhancing caregiver workforce sustainability through its unified system which detects problems and explains them while providing users with options to take action.

This research aligns with Sustainable Development Goal 3: Good Health and Well-being by addressing caregiver workforce sustainability, which is essential for delivering safe, high-quality healthcare services. A stable and well-supported caregiver workforce directly contributes to improved patient outcomes, reduced errors, and overall system resilience [14]. At the same time, the research aligns with the Centre of Excellence for Artificial Intelligence (CEAI) by applying advanced AI and data science techniques to solve a critical real-world healthcare problem. The proposed system integrates machine learning, temporal modeling, graph-based relational analysis, and explainable AI within a practical web-based solution, demonstrating how intelligent technologies can be used to enhance workforce management, support decision-making, and improve healthcare delivery in a scalable and impactful manner.

1.2. Literature Review

Recent scholarly research has moved beyond simply identifying caregiver burnout toward understanding how workload patterns, system interactions, and care environments collectively influence caregiver performance over time. For instance, Aiken et al. demonstrated that nurse workload and staffing ratios are strongly associated with both caregiver burnout and patient mortality, reinforcing that caregiver well-being is tightly linked to system-level outcomes rather than individual conditions alone [15]. Similarly, McHugh et al. found that hospital work environments, including staffing adequacy and managerial support, significantly affect nurse burnout and job outcomes, highlighting the importance of organizational context in shaping caregiver experience [16].

More recent studies have also explored how digital healthcare systems contribute to workload complexity. Asgari et al. reviewed the impact of Electronic Health Record (EHR) usage on clinicians and found that increased documentation requirements and system interaction lead to higher cognitive load and fatigue [17]. In addition, DiAngi et al. showed that patterns of EHR usage, such as after-hours documentation, can be used as indicators of workload stress and

inefficiency [18]. These studies indicate a shift toward understanding caregiver workload as a combination of clinical, administrative, and digital factors, although most research remains focused on analysis rather than system-level intervention.

Existing research approaches for analyzing caregiver workload and burnout can be broadly categorized into observational studies, predictive modeling, and operational optimization techniques. Observational studies focus on identifying relationships between workload conditions and caregiver outcomes. While these studies provide strong empirical evidence, they are largely retrospective, making them less suitable for real-time monitoring or proactive intervention [15], [16].

To address this limitation, predictive modeling approaches have been introduced. For example, Sendak et al. emphasized the role of machine learning in healthcare decision-making, particularly in identifying patterns from large-scale operational data [19]. Similarly, Rajkomar et al. demonstrated how deep learning models can be applied to healthcare datasets for predictive analysis, enabling more accurate identification of risk patterns [20]. However, these approaches are typically designed for clinical outcome prediction, and their application to caregiver workforce management remains limited.

Operational optimization approaches, such as nurse scheduling and staffing models, focus on improving resource allocation. These methods are effective in balancing workload constraints and ensuring adequate staffing, but they are generally designed from an efficiency perspective rather than a human-centered perspective. As a result, they do not explicitly account for behavioral factors such as fatigue accumulation, recovery patterns, or workload imbalance over time. This highlights a gap between predictive analytics and operational decision-making in existing research.

In real-world caregiving environments, systems such as Epic and Cerner EHR platforms play a central role in managing clinical and operational data. These systems provide dashboards that display staffing levels, patient assignments, and workload-related metrics, allowing administrators to monitor workforce conditions in real time [9], [10]. While these platforms have significantly improved data accessibility and coordination, their primary function remains administrative rather than analytical.

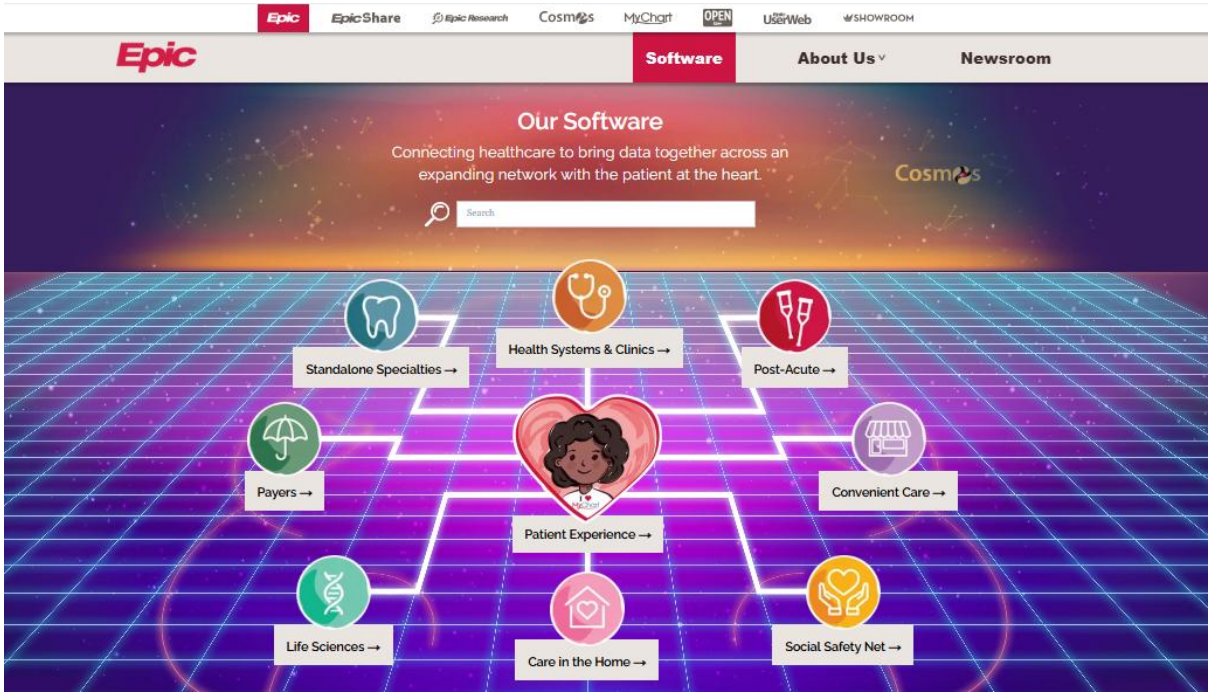


Figure 2: Example of EHR-Based Workforce Dashboard (Epic Systems, 2023)

Workforce management systems built on top of these platforms support scheduling, shift allocation, and resource planning. However, these systems largely rely on predefined rules and historical data, without incorporating predictive intelligence or behavioral analysis. As a result, they are limited in their ability to detect early warning signs of caregiver deterioration.

Emerging AI-based solutions have introduced predictive analytics into healthcare operations, aiming to improve efficiency and decision-making. Jiang et al. highlighted the potential of AI in healthcare for predictive modeling and automation, particularly in handling large-scale data [21]. However, these solutions are often focused on clinical outcomes or operational efficiency, rather than caregiver well-being. Additionally, they are typically implemented as standalone modules, lacking integration across different aspects of workforce management.

Despite significant advancements in both research and practical healthcare systems, several critical limitations remain. First, many existing approaches rely on static or short-term data, such as periodic surveys or snapshot-based operational metrics. While these provide useful insights, they fail to capture the dynamic and evolving nature of caregiver workload and deterioration over time. Even machine learning-based studies, such as those using EHR activity logs, often focus on short-term prediction rather than modeling longitudinal patterns of workload accumulation and recovery [18], [19].

Second, most existing systems treat caregivers as independent entities, without considering the relational and structural context in which caregiving occurs. In real-world environments, workload is shaped by team interactions, shared responsibilities, shift structures, and organizational dynamics. However, current models rarely incorporate these relational dependencies, resulting in an incomplete understanding of caregiver workload and its contributing factors [15], [16].

Third, there is a lack of integration across system components. Burnout assessment tools (e.g., MBI, ZBI), predictive models, and workforce management systems operate as isolated solutions, each addressing a specific aspect of the problem. This fragmentation prevents healthcare organizations from developing a holistic view of workforce conditions and limits their ability to make coordinated, data-driven decisions that balance operational efficiency with caregiver well-being [19], [20].

Finally, most existing systems focus primarily on monitoring or prediction, rather than supporting actionable intervention. While identifying burnout risk or workload imbalance is valuable, these systems rarely provide guidance on what actions should be taken next. This highlights a critical gap in current research and systems, where detection is not effectively linked to intervention, and emphasizes the need for integrated solutions that combine temporal analysis, relational modeling, and decision-support capabilities [20], [21].

Study / System	Approach Type	Data Used	Key Contribution	Limitation
Aiken et al. [15]	Observational	Staffing data	Links workload to burnout	No real-time detection
McHugh et al. [16]	Observational	Survey data	Work environment impact	Static analysis
Linzer et al. [11]	ML Prediction	Survey + admin	Predict burnout	No temporal modeling
HiPAL [12]	Deep Learning	EHR logs	Behavioral modeling	No intervention
Epic Systems [9]	Dashboard	Operational data	Real-time visibility	No predictive capability
Burke et al. [13]	Optimization	Scheduling data	Workforce allocation	Ignores human factors

Table 2: Comparison of Existing Approaches

1.3. Research Gap

Extensive research on caregiver burnout and workload has been conducted, yet the lack of systems to model how caregivers deteriorate as a process that occurs over time in continuous, evolving ways is clear. The majority of previous research on caregiver burnout has focused on identifying prevalence, associated factors, or providing short-term predictions about levels of burnout, but has not addressed the impact of workload patterns, scheduling structures, and various environmental exposure factors, all of which develop over time. For example, studies such as Dall'Ora et al. and Shah et al. highlight the relationship between long shifts, workload, and burnout, yet they remain descriptive and retrospective in nature [1], [3].

Similarly, existing digital systems such as Electronic Health Record (EHR) platforms and workforce dashboards provide operational visibility but lack the ability to model behavioral change, temporal progression, and cumulative workload impact [9], [10]. While machine learning approaches, including models proposed by Linzer et al. and Wang et al., attempt to predict burnout, they still treat the problem as a classification or prediction task rather than a dynamic process [11], [12]. Therefore, what is fundamentally missing is a unified, process-oriented system that integrates behavioral, temporal, and operational dimensions of caregiver workload into a continuous monitoring and intervention framework.

Current approaches are insufficient because they fail to address the multi-dimensional and interconnected nature of caregiver workload and deterioration. First, most systems rely on static indicators, such as survey scores or snapshot-based metrics, which do not reflect how workload accumulates or how recovery occurs over time. As highlighted in EHR-related studies, caregiver workload is not constant but fluctuates based on task complexity, documentation burden, and environmental factors [17], [18].

1.3.1. Novelty

The proposed research introduces a new way of thinking about caregiver workforce deterioration. Instead of treating it as a simple outcome, such as a stress score or burnout label, this research views deterioration as something that develops gradually over time. In real caregiving environments, no caregiver becomes overwhelmed in a single shift. It is the result of repeated exposure to demanding schedules, high workload, and limited recovery. By modeling deterioration as a dynamic process, the system can detect early warning signs before the situation becomes critical.

Another important contribution is the idea of living behavioral profiles. In reality, every caregiver is different. Some may handle high workloads better, while others may be more affected by irregular schedules or emergency situations. Existing systems often apply the same threshold to everyone, which is not realistic. This research instead builds a personal baseline for each caregiver, learning what is normal for them. This makes the system more accurate because it does not compare caregivers to others, but rather to their own typical patterns.

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Finally, and most importantly, the research moves beyond detection to actual intervention. Existing systems often stop at identifying risk, but do not support action. This system helps supervisors by suggesting practical workload redistribution strategies, ensuring that work is reassigned in a fair and balanced way. Additionally, because real-world data can be noisy and incomplete, the system incorporates noise-robust learning techniques such as co-teaching, making it more reliable in real healthcare environments.

Overall, the novelty of this research lies not in a single technique, but in how multiple advanced concepts are combined into a cohesive, real-world solution that detects, understands, and responds to caregiver workforce deterioration in a meaningful and actionable way.

The need for such an integrated system is strongly supported by existing evidence. Research consistently shows that caregiver burnout is influenced by workload, staffing levels, and organizational conditions, and is directly linked to patient safety and healthcare outcomes [6], [15]. Furthermore, studies indicate that digital systems, while improving operational efficiency, also introduce additional workload and cognitive burden, emphasizing the complexity of modern caregiving environments [17], [18].

At the same time, advancements in AI and machine learning demonstrate the feasibility of analyzing large-scale healthcare data for predictive insights [20], [21]. However, as highlighted in current literature, these technologies have not yet been effectively integrated into systems that support continuous monitoring, relational analysis, and actionable intervention.

Therefore, there is a clear and well-justified research gap in developing a system that combines behavioral modeling, temporal analysis, relational intelligence, and decision support into a unified solution. Addressing this gap is essential for improving caregiver well-being, reducing turnover, and enhancing overall healthcare system performance.

2. OBJECTIVE

2.1. Main Objective

The main objective of this research is to design and develop a web-based AI-assisted system that detects early patterns of caregiver workforce deterioration and supports adaptive workload redistribution using behavioral, temporal, and relational data analysis within a defined evaluation period.

This objective is defined using the SMART criteria as it specifically focuses on detecting caregiver workforce deterioration and enabling adaptive workload redistribution. It is measurable through evaluation metrics such as prediction accuracy, pattern detection capability, and the effectiveness of decision-support outputs. The objective is achievable as it is implemented as a web-based application using available operational and behavioral data. It is relevant because it directly addresses critical issues such as caregiver burnout, workload imbalance, and workforce sustainability identified in prior studies [3], [6]. Furthermore, it is time-bound, as the system is designed, developed, and evaluated within the defined project timeline.

2.2. Specific Objectives

Objective 1: Develop caregiver workload and behavioral feature modeling

This objective focuses on identifying and computing key caregiver-related indicators such as workload intensity, schedule stability, environmental strain, and self-reported experience using operational and behavioral data. It is measurable through quantifiable metrics including workload index, recovery time, and strain levels. The objective is relevant as prior studies have established strong links between workload conditions and caregiver burnout [1], [6]. The expected outcome is a structured set of features that accurately represent caregiver working conditions and form the foundation for further analysis.

Objective 2: Build dynamic personalized behavioral profiles

This objective aims to develop a system that learns and maintains a dynamic baseline for each caregiver, enabling detection of deviations from individual normal patterns rather than applying generalized thresholds. It is measurable through deviation scores that quantify how current behavior differs from baseline patterns. This objective is relevant as it addresses the limitations of standardized burnout measures that fail to capture individual variability [7]. The outcome is the creation of adaptive, individualized behavioral profiles that improve detection accuracy.

Objective 3: Implement temporal pattern detection and risk accumulation modeling

This objective focuses on analyzing sequential workload patterns and implementing a risk memory mechanism that captures how workload pressure accumulates and recovers over time. It is measurable through temporal trend detection accuracy and the modeling of risk accumulation curves. This objective is relevant as it overcomes the limitations of snapshot-based analysis in existing systems [12]. The outcome is a time-aware system capable of identifying gradual deterioration patterns rather than isolated events.

Objective 4: Model relational workforce structures using graph-based analysis

This objective aims to capture relationships between caregivers, teams, shifts, and environments using graph-based modeling techniques to identify workload imbalance and structural risk factors. It is measurable through the detection of clustering patterns and imbalance within teams and work assignments. This objective is relevant as research highlights the impact of staffing structures and team dynamics on caregiver outcomes [15]. The outcome is the generation of structural insights into workforce distribution and relational workload patterns.

Objective 5: Develop explainable decision-support and case narrative generation

This objective focuses on generating interpretable outputs for supervisors, including explanations of detected risks, contributing factors, and recommended actions. It is measurable through the clarity, usability, and interpretability of the generated insights. This objective is relevant as there is a growing need for explainable AI in healthcare decision-making systems [19], [21]. The outcome is a transparent decision-support interface that enables informed and practical decision-making.

Objective 6: Design and evaluate adaptive workload redistribution strategies

This objective aims to develop and validate mechanisms that support fair and effective workload redistribution based on detected risk patterns and caregiver compatibility. It is measurable through improvements in workload balance and reduction in high-risk cases after intervention. This objective is relevant as it addresses the gap between prediction and actionable intervention in existing systems. The outcome is a practical and implementable intervention strategy that supports caregiver well-being and workforce sustainability.

3. METHODOLOGY

This research adopts a DSR approach, as it aims to design, develop, and evaluate a practical solution to a real-world problem in caregiver workforce management. Design Science is appropriate because this study goes beyond analysis and proposes the development of an intelligent system artifact for detecting caregiver deterioration and supporting workload redistribution.

The research will follow iterative stages including problem identification, system design, development, evaluation, and refinement. In addition, elements of an experimental approach will be incorporated during the evaluation phase, where multiple analytical techniques will be tested and compared to determine their effectiveness in detecting workload patterns and deterioration.

The proposed system will apply a combination of data science and machine learning techniques to model caregiver workload and deterioration patterns. Initially, feature engineering techniques will be used to extract meaningful indicators such as workload intensity, schedule stability, recovery time, and environmental strain from raw operational data.

To analyze patterns, both supervised and unsupervised learning methods will be explored, including algorithms such as logistic regression, decision trees, and clustering techniques to identify workload patterns and categorize deterioration types. To capture temporal dependencies, time-series or sequence-based will be considered to analyze how workload evolves across shifts.

Furthermore, graph-based modeling techniques will be utilized to represent relationships between caregivers, teams, and shifts, enabling analysis of structural workload imbalance. A risk accumulation model will be designed to simulate how workload pressure builds and decays over time. To address the challenges of noisy or imperfect data, noise-robust learning methods such as co-teaching may be incorporated. The final selection of techniques will be determined based on experimental evaluation.

The system is proposed to be developed as a web-based application, ensuring accessibility for supervisors and stakeholders. The backend will be implemented using Python, due to its strong ecosystem for data analysis and machine learning, including libraries such as Pandas, NumPy, and Scikit-learn. For advanced modeling, frameworks such as TensorFlow or PyTorch may be utilized.

The frontend will be developed using modern web technologies, which is React to support interactive dashboards and visualization of insights. A suitable database system will be selected based on the type, scale, and structure of the collected data.

For visualization, tools such as Matplotlib or Plotly will be used to present trends, workload patterns, and risk indicators in an understandable format. These tools are selected due to their scalability, compatibility, and widespread use in data science applications, making them suitable for both development and potential future deployment.

This research will utilize a combination of primary, secondary, and synthetic data sources. Primary data may be collected through structured surveys or simulated caregiver inputs,

including self-reported workload perception and shift experience. Secondary data may include publicly available healthcare datasets and patterns identified from existing literature.

In situations where access to real-world data is limited due to ethical or institutional constraints, synthetic datasets will be generated based on realistic scenarios derived from domain knowledge and existing research findings. This approach will enable system development and evaluation while maintaining practical relevance.

Ethical considerations will be strictly followed throughout the research. Any data involving human participants will require informed consent, anonymization, and secure data handling practices. Sensitive information will not be stored in identifiable form, and the system will be designed to ensure privacy, confidentiality, and responsible use of data.

The proposed system will be evaluated using both technical performance metrics and practical usability measures. For predictive and analytical components, metrics such as accuracy, precision, recall, and F1-score will be used to assess model performance. For clustering and pattern detection, evaluation metrics such as silhouette score and cluster quality measures will be applied.

Temporal analysis methods will be evaluated based on their ability to identify meaningful workload trends and deterioration patterns over time. The effectiveness of the risk accumulation model will be assessed by its ability to distinguish between temporary workload fluctuations and sustained deterioration.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

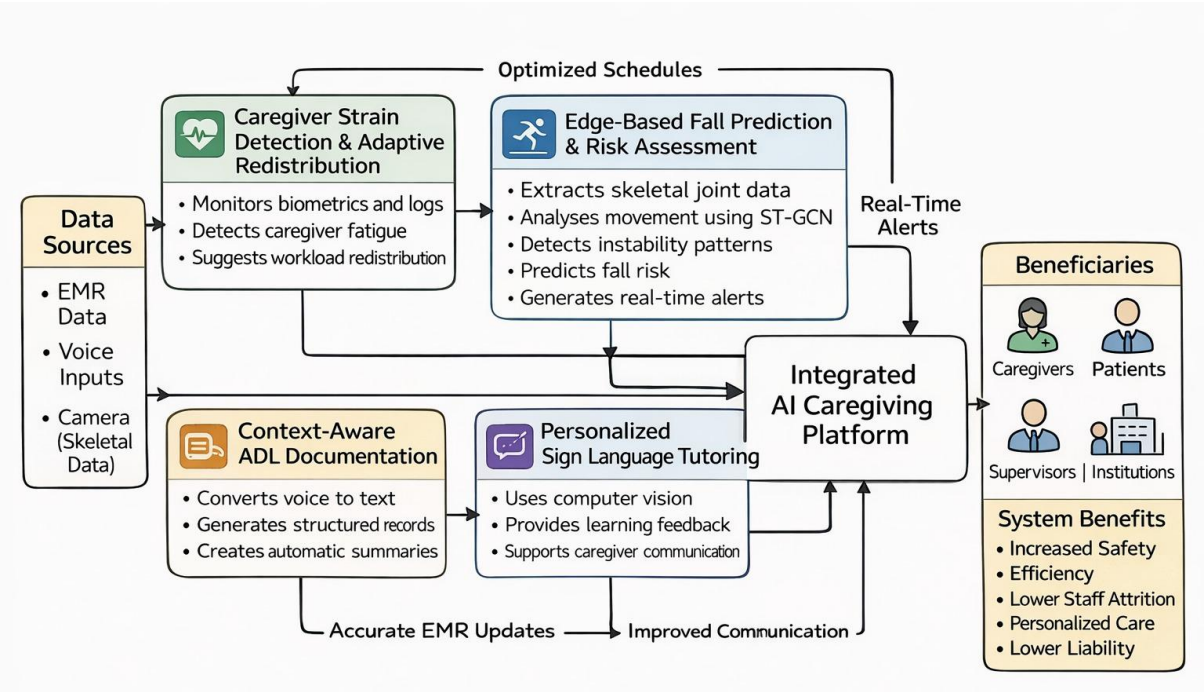


Figure 3: Overall Project System Overview Diagram

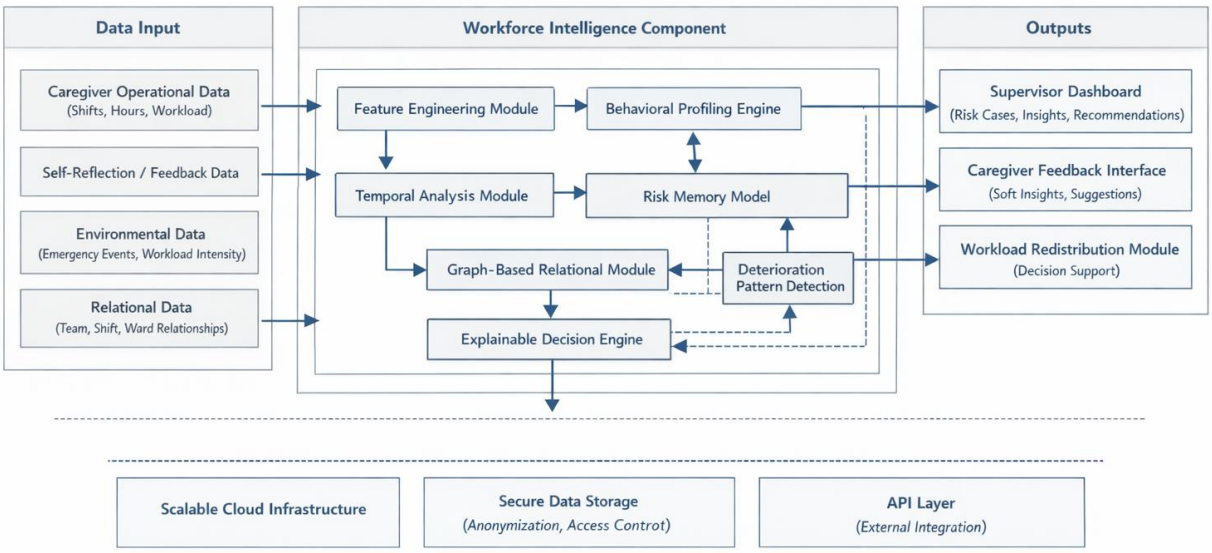


Figure 4: System Diagram for AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

The system diagram highlights the Workforce Intelligence Component as the core of the solution, consisting of modules such as Feature Engineering, Behavioral Profiling, Temporal Analysis, Risk Memory Modeling, Graph-Based Relational Analysis, Deterioration Pattern Detection, and an Explainable Decision Engine. Each module plays a specific role in transforming raw caregiver data into meaningful insights, enabling personalized analysis and early detection of deterioration patterns.

The diagram also illustrates the data flow and component interactions, where data from multiple sources such as operational records, self-reflection logs, environmental factors, and relational information is processed through interconnected modules. The flow shows how features are extracted, analyzed over time, and combined with relational context to detect risk

patterns, which are then translated into actionable outputs such as supervisor dashboards, caregiver feedback, and workload redistribution recommendations. Additionally, the diagram incorporates scalability and security considerations, ensuring real-world applicability.

5. USER REQUIREMENTS

The proposed system involves multiple stakeholders who interact with or are impacted by caregiver workforce management. The primary users of the system are caregivers, supervisors, and healthcare administrators. Caregivers interact with the system indirectly through data inputs such as self-reflection logs or workload-related information, and they benefit from personalized insights that support their well-being and work patterns. Supervisors act as key decision-makers, using the system to monitor workforce conditions, identify high-risk cases, and take appropriate actions such as workload redistribution.

Healthcare administrators and organizational stakeholders are also important users, as they rely on aggregated insights to support workforce planning and resource allocation. Additionally, patients and families are indirect stakeholders, as improvements in caregiver workload management directly contribute to better care quality and safety.

5.1. Functional Requirements

The system is expected to provide a set of core functionalities that support caregiver monitoring, analysis, and intervention. These functional requirements define what the system should do.

1. The system shall collect and store caregiver-related data, including workload, shift schedules, and self-reported experiences.
2. The system shall compute workload and behavioral indicators, such as workload intensity, recovery time, and environmental strain.
3. The system shall maintain personalized behavioral profiles for each caregiver and update them dynamically based on new data.
4. The system shall detect patterns of caregiver deterioration using temporal analysis and risk accumulation modeling.
5. The system shall analyze relationships between caregivers, teams, and shifts to identify workload imbalance and structural risks.
6. The system shall generate explainable case summaries, providing clear insights into detected risks and contributing factors.
7. The system shall provide decision-support recommendations for supervisors, including potential workload redistribution options.
8. The system shall visualize workload trends and risk indicators through an interactive dashboard.
9. The system shall allow supervisors to explore historical data and patterns for informed decision-making.

10. The system shall support scenario-based evaluation of workload redistribution strategies before implementation.

5.2. Non-Functional Requirements

In addition to functional capabilities, the system must satisfy several non-functional requirements to ensure usability, reliability, and practical deployment.

The system must ensure data security and privacy, particularly because caregiver-related data may include sensitive information. This includes implementing data anonymization, secure storage, and controlled access mechanisms.

The system should provide high performance and responsiveness, enabling real-time or near real-time analysis of workload data and generation of insights without significant delays.

Usability is a critical requirement, as the system is intended for non-technical users such as supervisors. Therefore, the interface must be intuitive, user-friendly, and easy to interpret, with clear visualizations and explanations.

The system should be scalable, allowing it to handle increasing amounts of data as more caregivers and operational records are incorporated.

Reliability is also essential, meaning the system should consistently produce accurate and stable outputs even when data is incomplete or noisy.

Finally, the system should maintain flexibility and adaptability, enabling updates to models, features, and data sources as the research progresses or as new requirements emerge.

6. COMMERCIALIZATION PLAN

6.1. Target market and customer persona

The target market for the proposed system includes care homes, rehabilitation centers, and healthcare organizations that manage caregiver workforce operations. The primary customer persona consists of healthcare administrators and supervisors responsible for staff scheduling, workload management, and operational efficiency. Secondary users include caregivers who indirectly benefit from improved workload balance and well-being support. The system is particularly valuable in environments with high workload intensity and staffing challenges, where proactive workforce management is critical.

6.2. Value proposition

The proposed system offers a unique value by providing an AI-assisted solution that not only detects caregiver workload deterioration early but also supports actionable intervention through workload redistribution. Unlike existing systems that focus on static monitoring or isolated predictions, this solution integrates behavioral, temporal, and relational intelligence to deliver meaningful insights. It helps organizations improve caregiver well-being, reduce burnout and turnover, and enhance patient care quality, making it both operationally and socially impactful.

6.3. Revenue model

The proposed system can be commercialized using a hybrid revenue model, combining both Software-as-a-Service (SaaS) and one-time deployment options. Under the SaaS model, healthcare organizations can subscribe to the platform on a monthly or annual basis, with pricing tiers based on the number of users, features, and analytics capabilities. In addition, the system can also be offered as a one-time licensed deployment, where the solution is installed within the client's infrastructure, particularly for organizations with strict data privacy or regulatory requirements. In this model, revenue can be generated through initial deployment fees, customization services, and ongoing maintenance or support contracts. This hybrid approach provides flexibility to accommodate different organizational needs, ranging from cloud-based scalability to on-premise deployment requirements.

6.4. Cost estimation

The proposed system offers a unique value by providing an AI-assisted solution that not only detects caregiver workload deterioration early but also supports actionable intervention through workload redistribution. Unlike existing systems that focus on static monitoring or isolated predictions, this solution integrates behavioral, temporal, and relational intelligence to deliver meaningful insights. It helps organizations improve caregiver well-being, reduce burnout and turnover, and enhance patient care quality, making it both operationally and socially impactful.

6.5. Competitive advantage

The key competitive advantage of the proposed system lies in its ability to integrate detection, explanation, and intervention within a single platform. Unlike traditional systems that provide only dashboards or predictions, this solution offers dynamic behavioral profiling, temporal pattern analysis, and workload redistribution support. Additionally, the system's explainable outputs and user-friendly design make it practical for real-world use by non-technical stakeholders.

6.6. Intellectual property considerations

The key competitive advantage of the proposed system lies in its ability to integrate detection, explanation, and intervention within a single platform. Unlike traditional systems that provide only dashboards or predictions, this solution offers dynamic behavioral profiling, temporal pattern analysis, and workload redistribution support. Additionally, the system's explainable outputs and user-friendly design make it practical for real-world use by non-technical stakeholders.

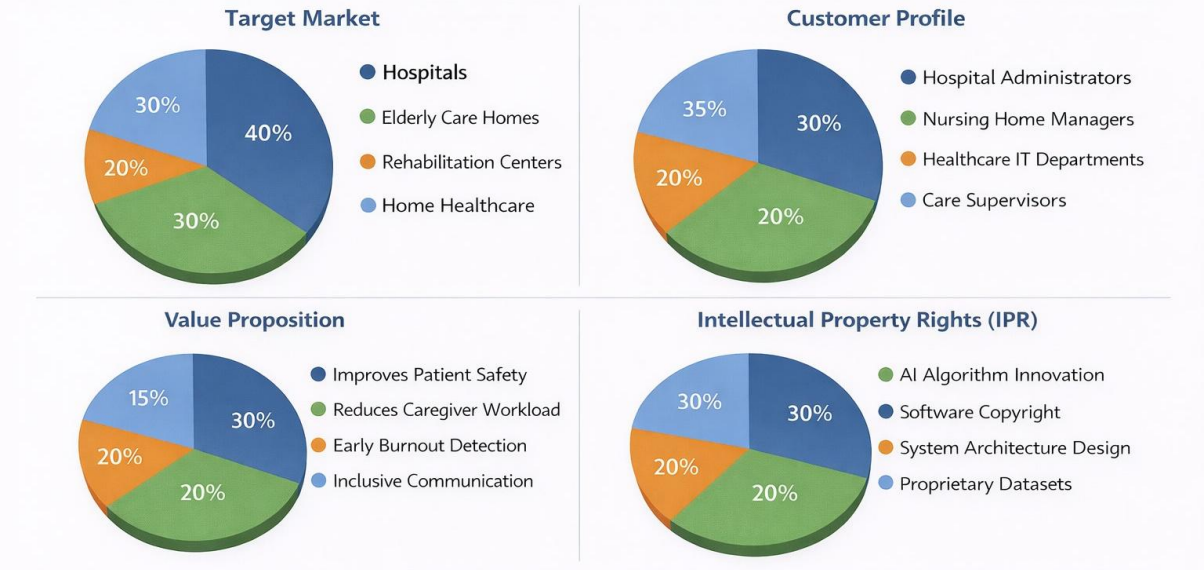


Figure 5: Commercialization Plan Factors related to overall project

7. BUDGET AND JUSTIFICATION

Category	Item	Estimated Cost (LKR)	Justification
Hardware	Backup Storage (External Drive)	25,000 – 50,000	Ensures secure storage and backup of datasets
Software Licenses	Development Tools	0 – 30,000	Mostly open-source; optional paid tools
Software Licenses	Design/Visualization Tools	0 – 15,000	Used for UI/UX design and visualization
Cloud Services	Cloud Hosting (AWS/GCP/Azure)	30,000 – 80,000	Supports deployment and remote access
Cloud Services	Data Storage & Compute	30,000 – 70,000	Supports model training and scalability
Human Effort	Development Effort	5,000- 10,000	Academic project work with human effort estimation
Human Effort	Expert Consultation	15,000 – 40,000	Optional: Domain validation and feedback

Table 3: Budget and Justification

Estimated Costs for Research Project

Description	Hours / Units	Cost (LKR)
Labour Costs		
Software Development	350 hrs	42,000
System Integration	250 hrs	30,000
Testing & Deployment	200 hrs	24,000
Total Labour Cost		**96,000**
Material & Infrastructure Costs		
USB Camera for Fall Detection	1 unit	12,000
Cloud Hosting & Storage (6 months)	–	10,000
Software Tools / APIs / GPU Usage (4 months)	–	8,000
Total Material & Infrastructure Cost		**30,000**
GRAND TOTAL		**126,000**

Figure 6: Cost Estimation Plan (draft)

8. WORK BREAKDOWN STRUCTURE (WBS)

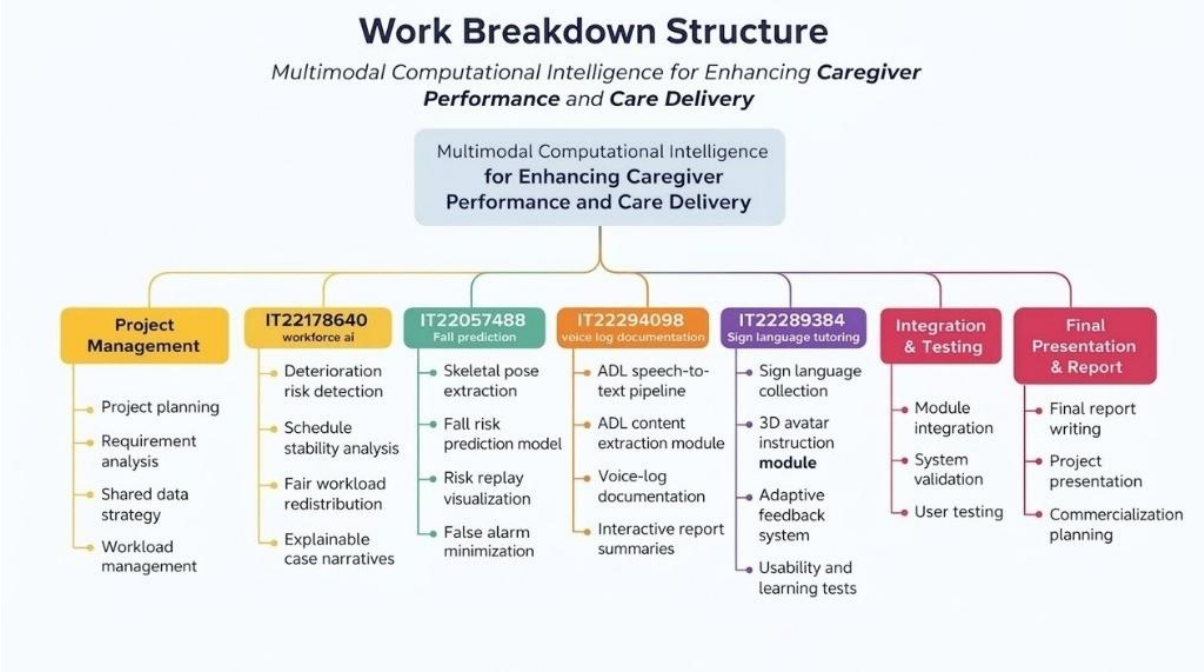


Figure 7: Work Break Down Structure

9. GANTT CHART

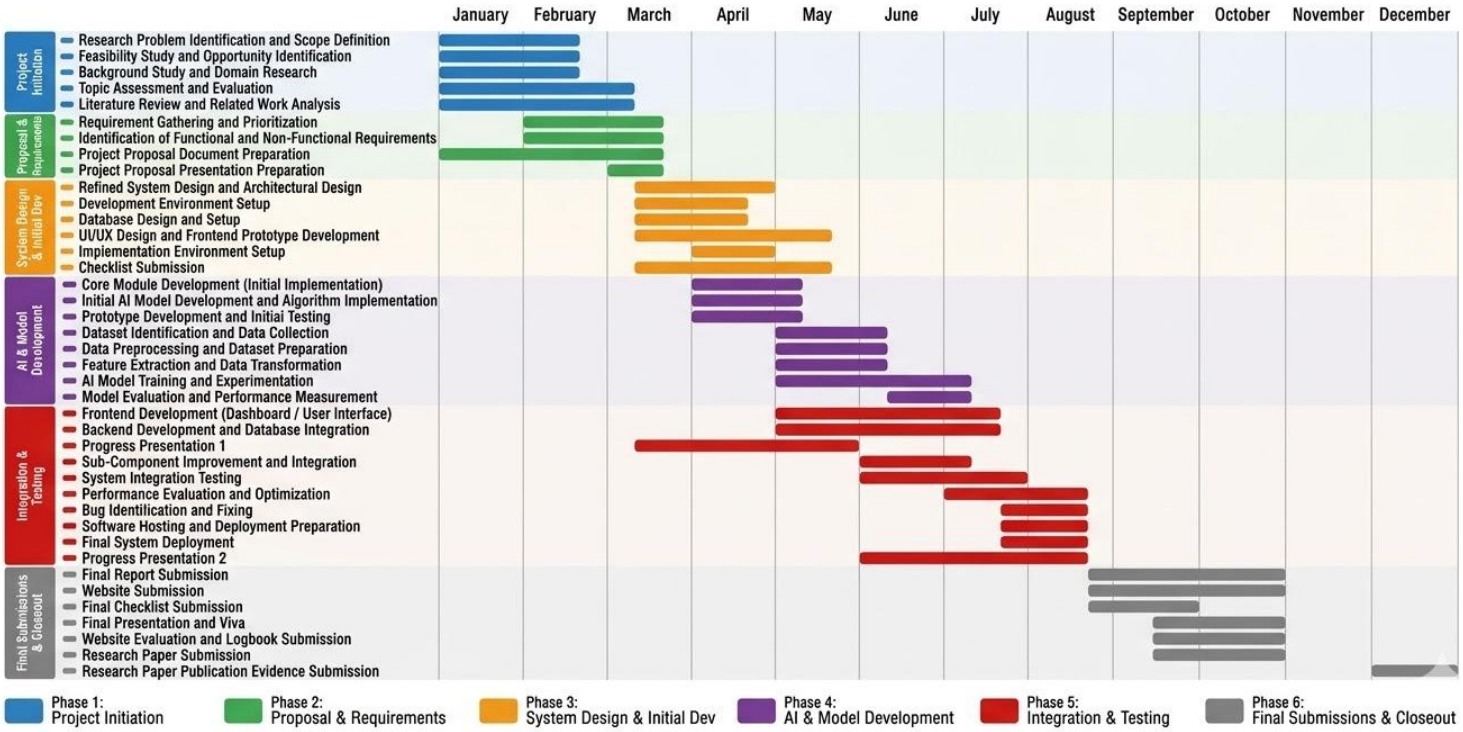


Figure 8: Gantt Chart

10.AI USAGE DISCLOSURE

AI tools were used in a limited and supportive capacity during the preparation of this research proposal. ChatGPT was utilized primarily to enhance language clarity, refine sentence structure, and assist in the organization of ideas. The purpose of using these tools was to improve the readability and presentation of the document, while ensuring that all core research concepts, problem formulation, and methodological decisions were developed independently by the author.

The extent of AI usage was strictly limited to minor writing assistance and idea refinement and did not involve generating full sections of the proposal or replacing original intellectual contribution. All AI-assisted outputs were critically reviewed, modified, and validated by the author to ensure accuracy, relevance, and alignment with academic standards. Validation methods included cross-checking information with credible academic sources, verifying technical correctness, and ensuring consistency with the research objectives. Full responsibility for the content presented in this proposal is retained by the author.

I hereby declare that:

1. I have read and understood the AI Usage Policy for the Research Project (IT4010).
2. I acknowledge that AI tools are to be used only as support tools, not as substitutes for my own understanding and work.
3. I agree to use AI tools responsibly, ethically, and transparently.
4. I will clearly disclose any use of AI tools in my submissions.
5. I understand that I bear full responsibility and accountability for any content, code, model, or output generated using AI tools.
6. I confirm that I will be able to explain, justify, and defend all aspects of my work during evaluations.
7. I understand that failure to comply with this policy may result in academic penalties.

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- Student ID: IT22178640
- Project Title: AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution
- Group ID : R26-DS-010

Signature

- Student Signature: 
- Date: 22.03.2026

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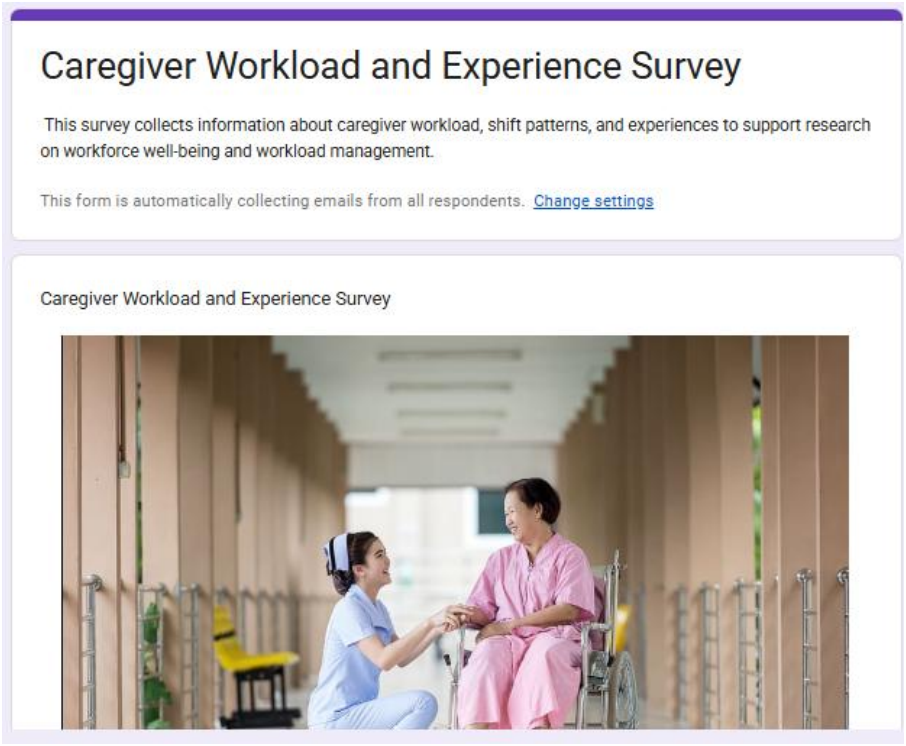
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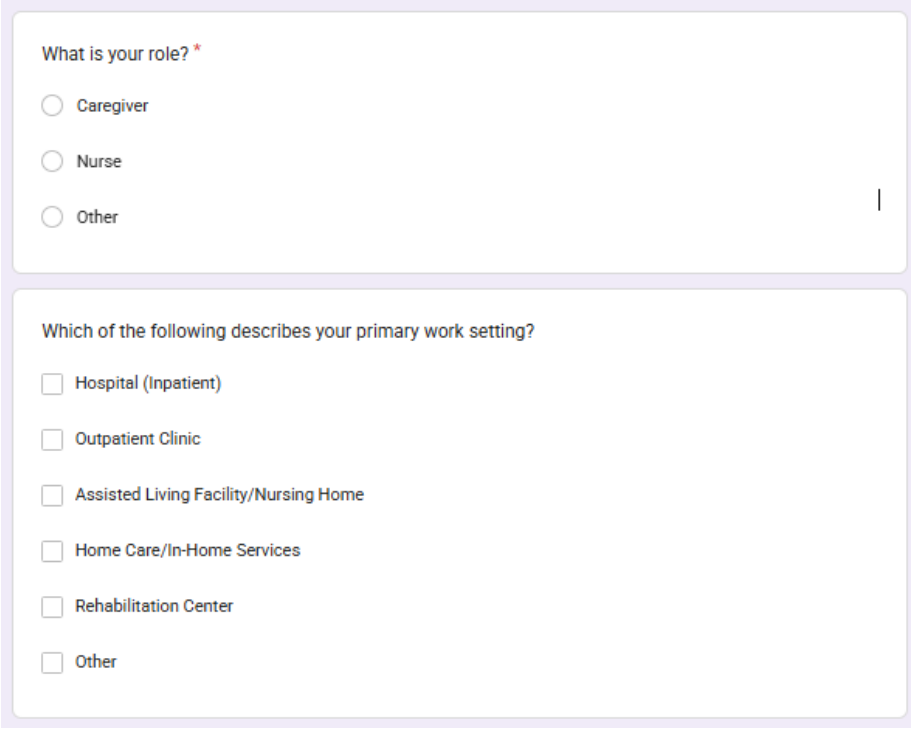
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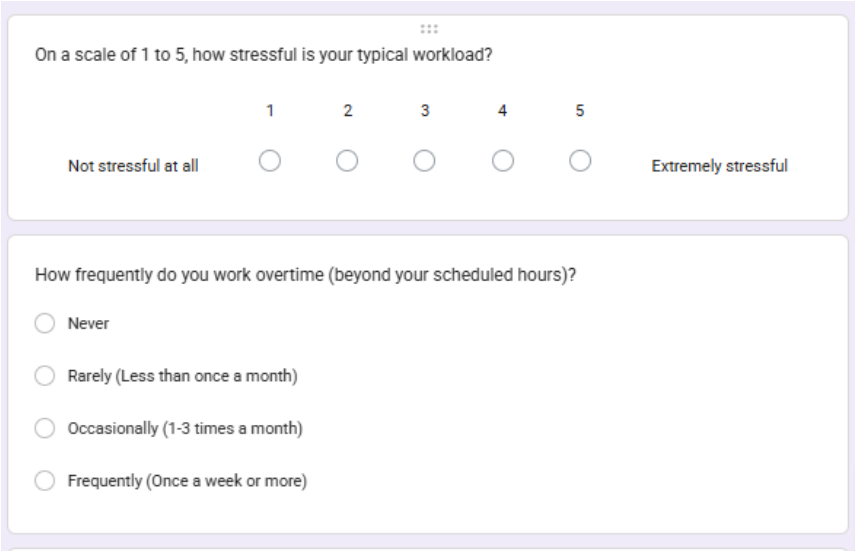
12.APPENDICES



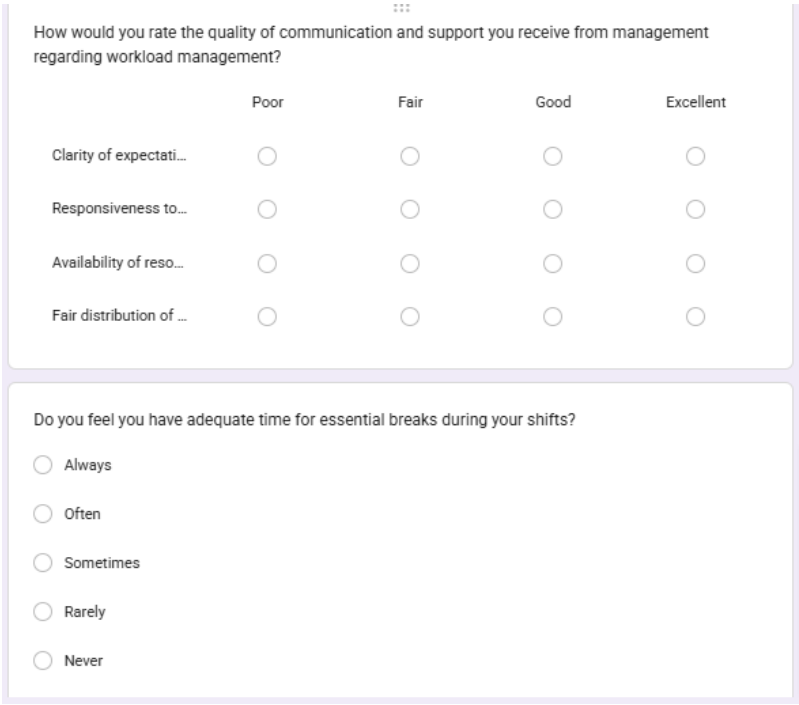
Appendix 1: Caregiver Workload and Experience Survey Front page



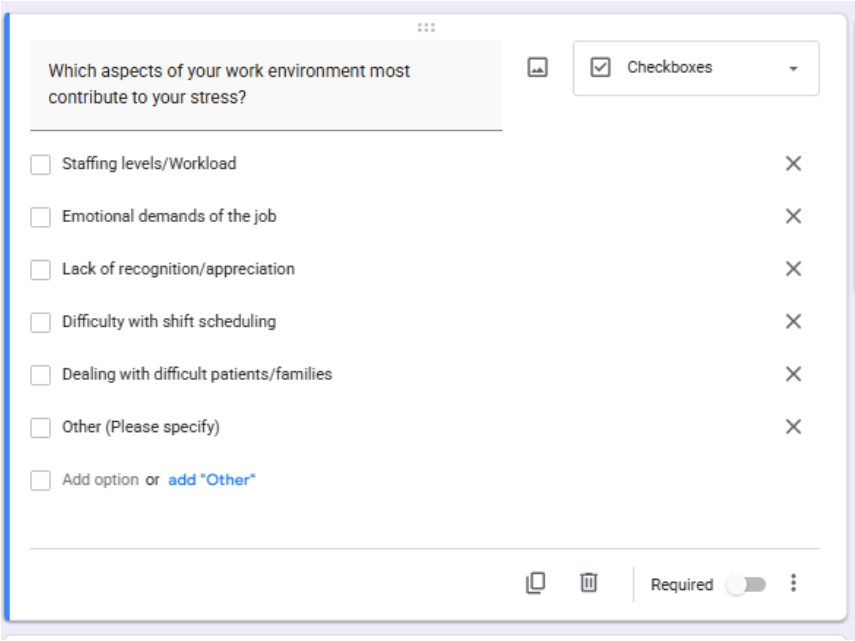
Appendix 2: Work setting questionnaire questions



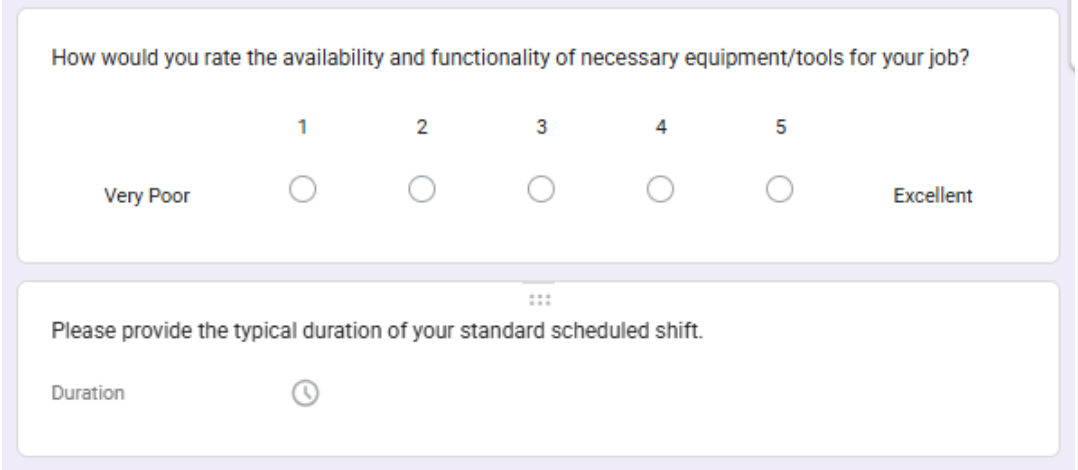
Appendix 3: Overtime and stress questionnaire questions



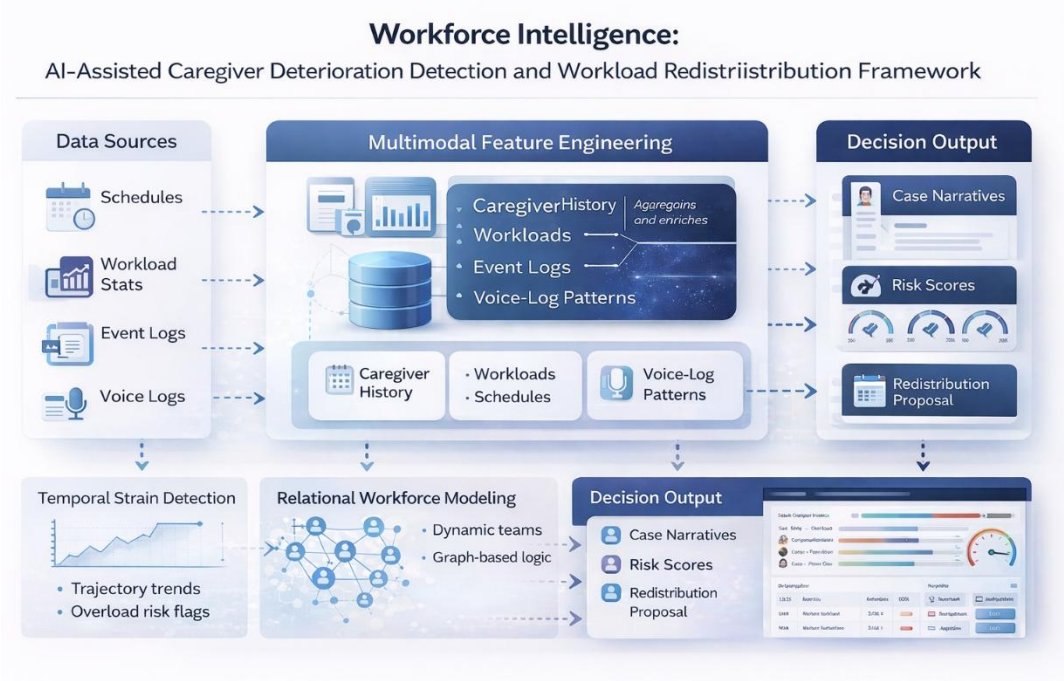
Appendix 4: Breaks and support questionnaire questions



Appendix 5: Stress factors questionnaire questions

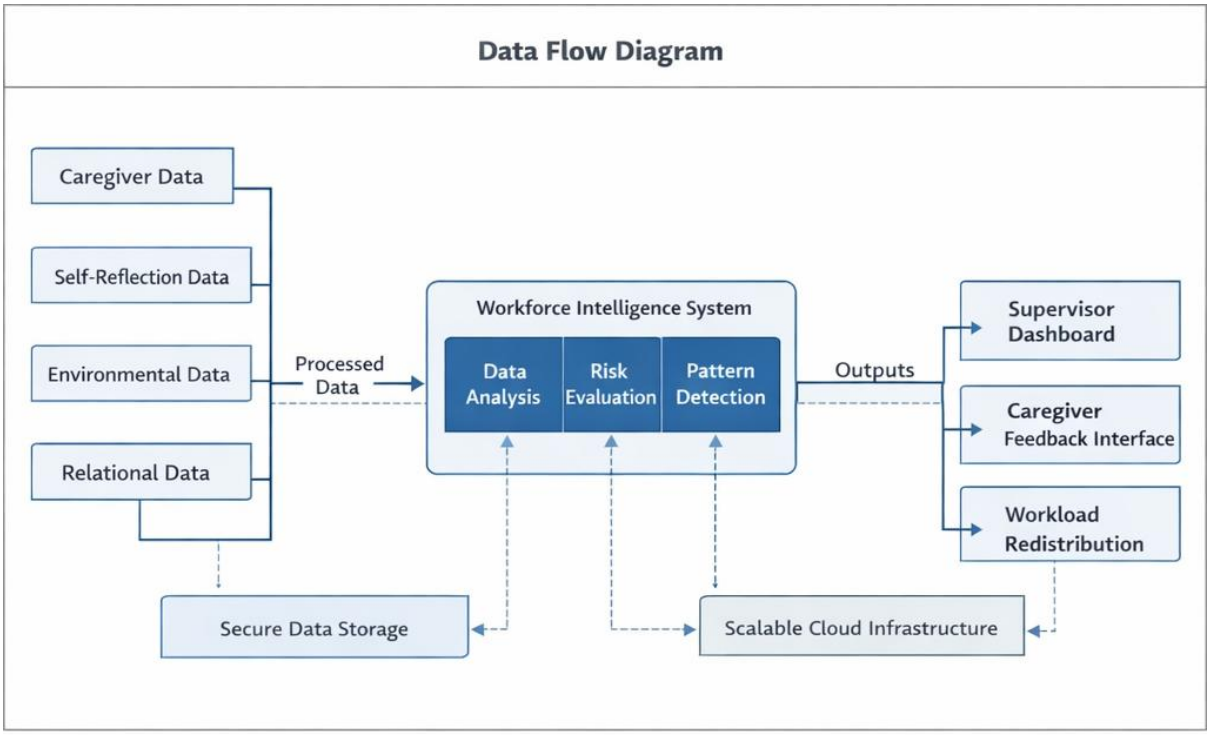


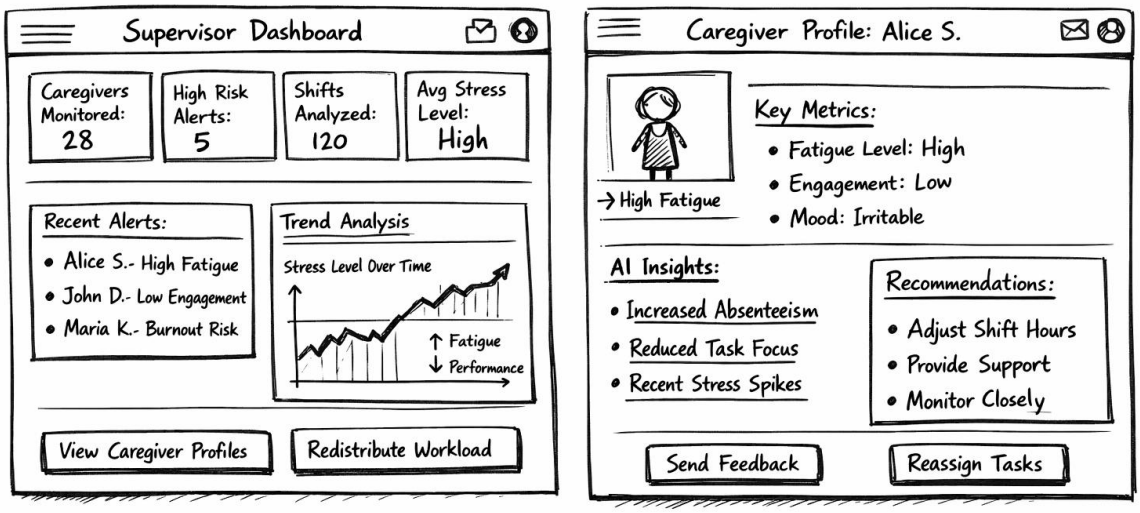
Appendix 6: Shift time questionnaire questions



Appendix 7: Component In-detailed overview diagram

Appendix 8: External Data Flow Diagram





Appendix 9: UI sample mockup